

SVM

↳ Popular algorithm used for classification problems.

Margin

↳ The distance between decision boundary and the closest data points.

→ The more higher is better

↳ classifies groups more accurately

→ support vectors

↳ specific data points are closest to the decision boundary

Decision Tree

↳ Supervised machine learning algorithm predicts outcomes by splitting data into branches.

entropy → measure of randomness with a dataset

information gain

↳ The measure of how much uncertainty is removed when splitting data.

Classification outputs $\begin{cases} \rightarrow \text{class labels} \\ \rightarrow \text{probabilities} \end{cases}$

class labels $\rightarrow$ predicting a direct category

probabilities $\rightarrow$ predicting the probability of a class.

Balanced Data sets

$\hookrightarrow$ When the number of target labels is relatively equal
uses accuracy metric

Imbalanced Data sets

$\hookrightarrow$ When there is a difference between classes
uses Recall, Precision
F-beta scores metrics.

The confusion matrix .

($\rightarrow$ 2x2 matrix used for binary classification that compares actual values against the model's predicted values.

True positive (TP) }
True Negative (TN) } accurate predictions

False positive (FP)

($\rightarrow$ model predicted a positive outcome }
The actual value was negative } type 1 Error

False negative

($\rightarrow$ model predicted a negative outcome }
but the actual value was positive } type 2 Error

precision

↪ positive prediction value

$$TP / (TP + FP)$$

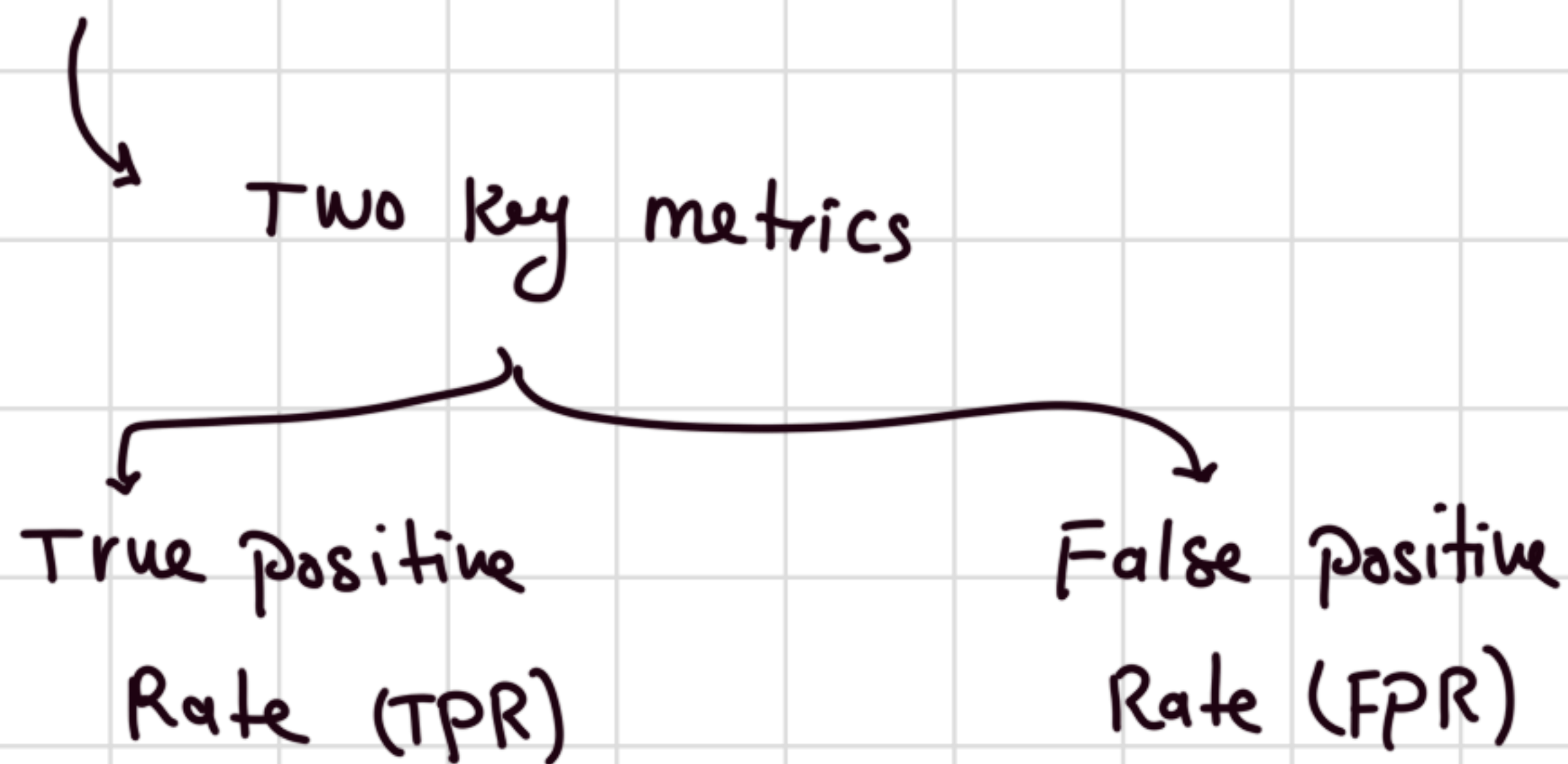
Recall

$$TP / (TP + FN)$$

Roc and AUC Curves

performance metrics primarily
used to evaluate binary
classification problems

Constructing Roc curve



AUC curve

The greater the area under the
curve, The better the model
is performing